Dresden University of Technology

Computational Logic ∴ Group

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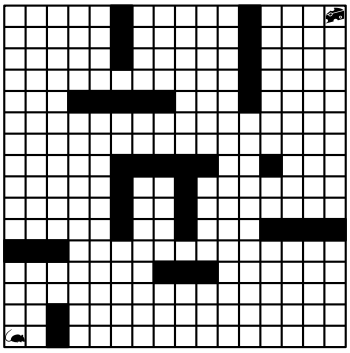
Advanced Problem Solving and Search

Winter Term 2026/27

Exercises 1

29/10/2026–2/11/2026

Problem 1.
Show how to use the A*-Algorithm, to find the shortest path in the maze below, starting from the mouse and ending in the cheese. The mouse can not use diagonal moves.



a) Which heuristics do you choose?

b) Which cells are explored, which not?

c) Which path does your A*-Algorithm use and is it really optimal?

d) How long is the shortest path?

Problem 2.
You are given an unlimited supply of coins with denominations from a set C, and a target amount n. Your task is to find the **minimum number of coins** needed to make exactly the amount n. If the amount cannot be made with the available denominations, report that it is impossible.

Instance for this exercise:

- Coin denominations: $C = \{1, 4, 6\}$
- Target amount: $n = 8$

Questions:

a) Design a dynamic programming algorithm to compute the minimum number of coins needed to reach amount n.

b) Trace the full DP table for the instance $C = \{1, 4, 6\}$, $n = 8$. State the optimal value and reconstruct the coins used.

c) What is the time and space complexity of your algorithm?

d) Compare your solution to a Greedy algorithm.

Problem 3.
Consider the *Bridge-Crossing Problem*, where 4 persons are on one side of a bridge and all of them need to get to the other side. It is nighttime and they have only one torch. However, the bridge is built to only withstand 2 persons. Also, the torch needs to be brought back to the remaining persons on the other side. Each person walks at a different speed and when they go together they must walk at the speed of the slowest. The goal is to find the minimal time for all persons to cross the bridge.

Person	Crossing Time
A	1 Min.
B	2 Min.
C	5 Min.
D	10 Min.

Design a dynamic programming algorithm to compute the solution.